

QUANTITATIVE CHEMISTRY – RELATIVE FORMULA MASS

KEY VOCABULARY

Relative atomic mass (A_r): the average mass of an atom compared to ^{12}C .

Relative formula mass (M_r): the sum of the A_r of all atoms in a formula.

Subscript: the small number showing how many atoms.

Bracket: multiplies everything inside it by the number after it.

COMMON MISCONCEPTIONS

~~M_r has units of grams.~~

✓ Relative masses have no units.

~~Ignore the number after a bracket.~~

✓ Multiply everything in the bracket by it.

~~M_r = number of atoms.~~

✓ M_r = sum of the atomic masses.

WORKED EXAMPLE: $\text{Ca}(\text{OH})_2$

A_r : Ca = 40, O = 16, H = 1.

Ca = 40; $(\text{OH})_2 = 2 \times (16 + 1)$.

= 40 + 2 × 17

$M_r = 40 + 34 = 74$

1

SIMPLE M_r

Calculate the M_r of CO_2 . (C=12, O=16) [1 mark]

Answer: _____ (M_r)

2

WATER

Calculate the M_r of H_2O . (H=1, O=16) [1 mark]

Answer: _____ (M_r)

3

USE BRACKETS

Calculate the M_r of $\text{Mg}(\text{NO}_3)_2$. (Mg=24, N=14, O=16) [2 marks]

Answer: _____ (M_r)

4

PERCENTAGE MASS

Calculate the % by mass of oxygen in CO_2 . [2 marks]

Answer: _____ %

5

FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

Relative formula mass is the _____ of the atomic masses and has _____ units.

sum

no

product

kg

6

FIND THE FORMULA

A compound CaCO_3 has $M_r = ?$. (Ca=40, C=12, O=16) Calculate it. [2 marks]

Answer: _____ (M_r)